Literature Status: Countable Discrete Extensions of Separable Compact Lines

Run `fa_banach_001`

23 July 2026

Status: `literature_already_answered (compact-line subproblem)`.

Source question

Korpalski and Plebanek ask in Section 7, Problem 7.1, on page 16 of [1] whether it is relatively consistent that, whenever K is a separable compact space of weight ω_1,

$$\eta(K, L) < \infty \qquad \text{for every } L \in \mathrm{CDE}(K).$$

Equivalently, every such countable discrete extension L should admit a bounded extension operator $C(K) \to C(L)$. Immediately after the problem, the authors explicitly ask whether it can be settled for separable compact lines.

Explicit later answer

Avilés and Korpalski prove the following as Theorem 1.4 on page 2 of [2]:

> Under $\mathsf{MA}(\kappa)$, if K is a separable compact line of weight κ, then for each countable discrete extension L of K there is an extension operator $E : C(K) \to C(L)$ of norm at most three.

Taking $\kappa = \omega_1$ gives the requested relative consistency statement for the entire compact-line class. The paragraph following Theorem 1.4 explicitly states that the theorem solves the compact-line problem posed in the source paper, so this is an author-identified answer rather than a new agent inference.

Scope limitation

Theorem 1.4 answers the compact-line subproblem completely and strengthens mere existence by giving $\|E\| \leq 3$. The supporting paper explicitly leaves open the broader question for arbitrary separable compact spaces.

Search evidence

A bounded search on 23 July 2026 used the exact source title together with the phrases "separable compact lines", "Martin's axiom", "countable discrete extension", and "extension operator". It located arXiv:2403.19327 and its open-access 2024 journal version.

References

[1] M. Korpalski and G. Plebanek, *Countable discrete extensions of compact lines*, Fund. Math. **265** (2024), 75–93; arXiv:2305.04565.

[2] A. Avilés and M. Korpalski, *Barely alternating real almost chains and extension operators for compact lines*, RACSAM **118** (2024), Paper 148; arXiv:2403.19327, doi:10.1007/s13398-024-01651-7.